

Behavioral Modeling of Urban Road Users

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Applications

Submitted by
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Spring Semester, 2024-25
April 23, 2025

DECLARATION

I certify that

- (a) I have completed the work on this report under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: April 23, 2025 (Kshamit Bombarde - 20BT3AI07) Place: Kharagpur

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This is to certify that the project report entitled “**Behaviour Modelling for Urban Road Users**” submitted by **Kshamit Bombarde** (Roll No. 20BT3AI07) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Dual Degree in Biotechnology and Biochemical Engineering / Artificial Intelligence and Machine Learning is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2024 - 25.

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ABSTRACT

Behavioral modeling of urban road users is increasingly essential in the context of evolving transportation systems, particularly with the rise of automated vehicles and smart city initiatives. Understanding how different road users—drivers, pedestrians, cyclists—interact within urban environments is crucial for designing systems that enhance safety, efficiency, and sustainability. Traditional models often fail to account for the complex and heterogeneous nature of urban traffic, which can lead to ineffective traffic management and increased accident rates. As urbanization accelerates, the need for accurate predictive models that can simulate real-world conditions becomes more pressing. These models help in anticipating user behavior, thereby informing infrastructure design and traffic regulations that cater to diverse user needs.

This research undertakes a comprehensive examination of urban traffic dynamics through multi-faceted behavioral modeling approaches. The study first explores existing datasets that capture the intricate interactions between pedestrians and vehicles in urban settings. Notable among these are the JAAD (Joint Attention for Autonomous Driving) and CARLA (Car Learning to Act) datasets, which offer detailed annotations of road user behaviors across diverse scenarios and environmental conditions. These datasets serve as valuable repositories of information regarding pedestrian crossing patterns, driver responses, and environmental influences on decision-making processes.

Through statistical analysis and pattern recognition techniques, the research aims to decode the complex interplay of factors that govern road user behavior. The methodology employs both quantitative metrics and qualitative behavioral markers to identify key determinants of traffic flow and safety. Of particular interest is the relationship between pedestrian crossing behaviors and driver responsiveness, especially in situations where formal infrastructure such as designated crosswalks is absent or ignored.

To validate theoretical findings and test hypotheses, the research utilizes SUMO (Simulation of Urban Mobility), a powerful traffic simulation platform capable of recreating complex urban environments with high fidelity. These simulations allow for systematic evaluation of how variations in pedestrian behavior—particularly arbitrary crossing patterns—impact overall traffic efficiency and safety. The research parameters are carefully calibrated to reflect realistic human behaviors, including reaction times, risk tolerance, and decision-making tendencies under various conditions.

The study specifically addresses two critical questions facing urban planners and transportation engineers: how arbitrary pedestrian crossings affect traffic efficiency, and to what extent such crossings contribute to unsafe vehicle-pedestrian interactions. By establishing a mathematical relationship between pedestrian density and traffic flow parameters, the research provides quantifiable insights into these complex dynamics. Furthermore, the conflict probability between vehicles and pedestrians is modeled to identify critical thresholds at which intervention becomes necessary.

Based on the analysis of both real-world data and simulation results, the research develops a comprehensive set of recommendations for urban infrastructure design and traffic management. These evidence-based suggestions span from physical infrastructure modifications to technological interventions and policy frameworks. The recommendations are stratified according to implementation complexity, cost considerations, and expected impact, providing urban planners with a practical toolkit for addressing pedestrian-vehicle conflicts.

The significance of this research extends beyond theoretical understanding to practical applications in urban design and intelligent transportation systems. By bridging the gap between behavioral science and traffic engineering, the study contributes to the development of more human-centered urban environments that balance efficiency with safety. The findings have particular relevance for emerging smart city initiatives and autonomous vehicle integration, where predictive modeling of human behavior represents a critical challenge.

This comprehensive approach to behavioral modeling offers valuable insights for enhancing the design of urban transportation systems, potentially reducing accidents and improving traffic flow while accommodating the needs of all road users in increasingly complex urban environments.

Chapter 1

1.1 Finding out existing pedestrian/driver datasets

Here's a detailed description of the datasets mentioned, including their key features, methodologies, and applications:

1. JAAD Dataset (Joint Attention in Autonomous Driving)

- Overview: The JAAD dataset studies pedestrian and driver behaviors at crosswalks, particularly the interactions that occur when pedestrians are about to cross the street.
- Content:
 - Video Clips: 346 HD video clips, each lasting 5-10 seconds, were recorded at 30 FPS.
 - Annotations:
 - We are bounding boxes with occlusion tags for all pedestrians.
 - Behavior annotations for each pedestrian (e.g., stopped, walking, looking).
 - Demographic attributes such as age and gender.
 - Environmental tags (weather conditions, traffic signals).
 - Total Annotations:
 - 378,643 bounding boxes across 2,786 pedestrians.
 - 82,032 total annotated frames.
- Purpose: Designed for research in pedestrian detection and understanding interactions at crosswalks to improve autonomous vehicle systems.

2. IUPUI-CSRC Pedestrian Situated Intent (PSI) Dataset

- Overview: This dataset aims to capture the dynamic intent changes of pedestrians and provide explanations of driver reasoning during interactions.
- Content:
 - Scenes: PSI-1.0 includes 110 unique scenes, while PSI-2.0 expands this to 196 scenes with additional annotations from new subjects.
 - Annotations:
 - Over 682,378 bounding box annotations for traffic objects and agents.
 - 71,259 unique pose estimations for pedestrians.
 - Text-based explanations of driver reasoning during pedestrian encounters.
- Innovative Features:
 - Dynamic intent changes for pedestrians identified by human drivers.
 - Integration of vision-language models for better understanding of interactions.
- Applications: Useful for developing algorithms that predict pedestrian behavior and improve vehicle-pedestrian interaction understanding.

3. PEDS Dataset

- Overview: The PEDS dataset is designed to study pedestrian behavior in urban environments through video recordings.
- Content:
 - Video footage capturing various pedestrian interactions in different urban settings.
 - Annotations include pedestrian movements and interactions with vehicles.
- Purpose: Focuses on understanding how pedestrians behave in real-world scenarios to aid in traffic safety research and autonomous vehicle development.

4. Oxford RobotCar Dataset

- Overview: A large-scale dataset consisting of driving data collected over a year in Oxford, UK, under varying weather conditions and times of day.
- Content:
 - Over 1000 km of driving data with extensive annotations for both vehicle and pedestrian behaviors.
 - Includes sensor data from LIDAR, cameras, and GPS systems to provide a comprehensive view of the driving environment.
- Purpose: Aimed at advancing research in autonomous driving systems by providing rich contextual data on urban driving scenarios.

5. Carla Driver Behavior Dataset

- Overview: This dataset is generated using a virtual simulator to capture various driving behaviors under controlled conditions.
- Content:
 - Data includes information on driver actions and responses during simulated driving scenarios.
 - Focuses on understanding driver behavior patterns through a virtual environment that mimics real-world situations.
- Purpose: Useful for training machine learning models aimed at predicting driver behavior in autonomous vehicles.

6. Driving Behavior Datasets from Kaggle

- Overview: These datasets focus on aggressive driving behaviors captured through accelerometer and gyroscope data from vehicles.
- Content:
 - Data includes instances of aggressive maneuvers such as speeding or sudden stops, recorded from various vehicles equipped with sensors.
- Purpose: Aimed at developing algorithms that can detect aggressive driving patterns to enhance road safety measures.

1.2 Techniques for Creating/Recording Datasets

Here's a detailed explanation of the techniques used to create and record pedestrian and driver behavior datasets, structured in pointers for clarity:

1. Onboard Cameras

- **Definition:** Cameras integrated into vehicles (cars, buses, trains) to capture real-time video footage of driving scenarios and pedestrian interactions.
- **Functionality:** These cameras can record high-definition videos and images from the vehicle's perspective, providing a first-person view of the environment.
- **Design Considerations:** Onboard cameras are designed to withstand harsh conditions, including vibrations, varying light conditions, and extreme temperatures.

2. Video Recording

- **High-Definition Footage:** Datasets like JAAD utilize HD video clips (5-10 seconds long) recorded at 30 frames per second (FPS), capturing detailed interactions between drivers and pedestrians⁵.
- **Diverse Conditions:** Recordings are made in various environments and weather conditions to ensure the dataset is comprehensive and representative of real-world scenarios.

3. Behavioral Annotations -

- **Timestamped Actions:** Each video frame is annotated with specific actions performed by pedestrians and drivers, such as walking, stopping, or looking. This allows for precise tracking of behaviors over time.
- **Demographic Data:** Annotations often include demographic attributes of pedestrians (e.g., age, gender) and contextual information (e.g., weather conditions, time of day) to enrich the dataset.

4. Bounding Box Annotations -

- **Object Detection:** Pedestrians are marked with bounding boxes in each frame to facilitate detection and tracking. For example, the JAAD dataset includes over 378,000 bounding boxes for pedestrians across its video clips⁵.
- **Occlusion Tags:** These tags indicate when pedestrians are obscured by other objects, which is crucial for training algorithms to handle real-world complexities.

5. Pose Estimation Techniques -

- **Analysis of Movement:** Advanced techniques are used to estimate the physical stance and movement patterns of pedestrians, providing insights into their behavior in relation to vehicles.
- **Integration with Video Data:** Pose estimation data can be combined with video footage to analyze how pedestrian movements correlate with driver reactions.

6. Sensor Fusion -

- **Multimodal Data Collection:** Some datasets integrate data from various sensors (e.g., LIDAR, radar) alongside video footage to provide a more comprehensive view of the environment.
- **Enhanced Contextual Understanding:** This approach helps in understanding complex interactions between vehicles and pedestrians by combining visual data with spatial awareness from other sensors.

7. Real-Time Data Processing -

- **Streaming Capabilities:** Onboard systems often include capabilities for real-time video streaming and processing, allowing for immediate analysis of driver-pedestrian interactions as they occur.
- **Data Storage Solutions:** Systems are equipped with network video recorders (NVRs) that store collected footage securely for later analysis.

8. Event Logging Software -

- **Behavior Tracking Tools:** Software like BORIS is used for logging events during video observations, allowing researchers to capture detailed behavioral data efficiently.
- **Structured Data Output:** The software produces structured output files containing timestamps and action labels for each subject in the video.

These techniques collectively enhance the quality and usability of datasets focused on pedestrian and driver behaviors, facilitating advancements in traffic safety research and the development of autonomous vehicle systems.

1.3 Key differences between object detection and action recognition in video datasets

Object detection focuses on identifying and localizing objects within individual frames, while action recognition analyzes sequences of frames to classify dynamic behaviors over time. The key differences between object detection and action recognition in video datasets can be summarized as follows:

1. Definition and Purpose

- **Object Detection:**
 - **Definition:** Object detection involves identifying and localizing specific objects within an image or video frame. It provides bounding boxes around detected objects along with their class labels.
 - **Purpose:** The primary goal is to recognize and locate objects of interest (e.g., cars, pedestrians) in a scene, enabling further analysis or interaction.
- **Action Recognition:**
 - **Definition:** Action recognition refers to the process of identifying specific actions or activities performed by subjects in a video sequence. This involves analyzing the temporal dynamics of movements over multiple frames.
 - **Purpose:** The goal is to classify what action is being performed (e.g., walking, running, jumping) based on the movement patterns observed in the video.

2. Input Data

- **Object Detection:**
 - Typically processes individual frames or images independently, focusing on identifying objects present in that specific frame.
 - Input can be a single image or a series of frames, but the analysis is often frame-by-frame without considering the temporal context.
- **Action Recognition:**
 - Requires sequences of frames (video clips) to analyze how actions unfold over time. It relies on motion information and changes between frames to classify actions.
 - Input data is inherently temporal, as it needs to capture the progression of movements that characterize specific actions.

3. Output

- **Object Detection:**
 - Outputs include bounding boxes around detected objects along with class labels for each object. For example, in a street scene, it might identify and label cars, pedestrians, and traffic signs.
- **Action Recognition:**
 - Outputs are typically class labels that describe the action being performed throughout the video segment. For instance, it may classify a sequence as "running" or "dancing" based on the observed movements.

4. Techniques Used

- Object Detection:
 - Common techniques include Convolutional Neural Networks (CNNs), Region-based CNNs (R-CNN), YOLO (You Only Look Once), and SSD (Single Shot MultiBox Detector). These algorithms focus on detecting and localizing objects efficiently within an image.
- Action Recognition:
 - Techniques often involve using CNNs for spatial feature extraction combined with Recurrent Neural Networks (RNNs) or Long Short-Term Memory Networks (LSTMs) to capture temporal dynamics. More recent approaches may use 3D CNNs that consider both spatial and temporal dimensions simultaneously.

5. Applications

- Object Detection:
 - Widely used in various applications such as autonomous driving (detecting pedestrians and vehicles), surveillance systems (monitoring activities), and retail analytics (customer behavior tracking).
- Action Recognition:
 - Commonly applied in security systems (detecting suspicious behaviors), sports analytics (analyzing player movements), and human-computer interaction (gesture recognition).

6. Complexity

- Object Detection:
 - Generally less complex than action recognition since it focuses on identifying static features within individual frames rather than analyzing dynamic movements over time.
- Action Recognition:
 - More complex due to the need for understanding motion patterns and sequences of actions, which requires sophisticated algorithms capable of processing temporal information effectively.

Chapter 2

2.1 Basic Statistics of JAAD and CARLA

In this section, we will explore the fundamental statistics of two key datasets relevant to our study: the JAAD (Joint Attention for Autonomous Driving) dataset and the CARLA (Car Learning to Act) dataset.

JAAD Dataset:

The JAAD (Joint Attention for Autonomous Driving) dataset is a pivotal resource designed to study pedestrian and driver behaviors at crosswalks, particularly focusing on interactions when pedestrians are about to cross the street. This dataset is instrumental in advancing research in pedestrian detection and understanding interactions at crosswalks, ultimately contributing to the development of safer autonomous vehicle systems.

Overview and Key Features:

- **Purpose:** The JAAD dataset aims to improve the understanding of pedestrian behavior in urban traffic environments, providing insights into how pedestrians interact with vehicles.
- **Dynamic Intent Changes:** It captures the dynamic intent changes of pedestrians and includes text-based explanations of driver reasoning during these interactions.
- **Comprehensive Annotations:** The dataset features extensive computer vision annotations across various scenes, making it an invaluable tool for developing algorithms that predict pedestrian behavior and enhance vehicle-pedestrian interaction understanding.

Basic Statistics:

- **Location of Data Collection:** The dataset was recorded in various urban settings across North America and Europe, ensuring a diverse representation of pedestrian behaviors.
- **Number of Videos/Frames:** The JAAD dataset comprises 346 high-definition video clips, each lasting between 5 to 10 seconds, totaling approximately 82,032 annotated frames.
- **Types of Vehicles:** The dataset primarily includes standard passenger vehicles involved in interactions with pedestrians.
- **Pedestrian Count:** A total of 2,786 pedestrians were captured in the videos, with behavior annotations available for 686 individuals.

CARLA Dataset:

The CARLA (Car Learning to Act) dataset is designed for autonomous driving research, featuring simulated urban environments that replicate real-world conditions. This dataset provides a rich set of data for training and evaluating algorithms in perception, decision-making, and control.

Overview and Key Features:

- **Purpose:** The CARLA dataset aims to enhance research in autonomous driving systems by providing comprehensive data on vehicle and pedestrian interactions within simulated environments.
- **High-Resolution Imagery:** Images are captured at resolutions up to 1920x1080 pixels, ensuring clarity for analysis.
- **Diverse Scenarios:** The dataset includes over 20 different weather conditions (sunny, rainy, foggy) and various times of day (day, night), allowing researchers to study driving behaviors under varying circumstances.
- **Comprehensive Annotations:** It features annotations for over 10,000 traffic signs, lane markings, and dynamic objects, contributing to realistic scenario modeling.

Basic Statistics:

- **Location of Data Collection:** The CARLA dataset is generated from simulated urban environments designed to reflect real-world conditions.
- **Number of Videos/Frames:** Approximately 10,000 video sequences are included in the dataset, resulting in over 1.5 million frames.
- **Types of Vehicles:** The dataset features five types of vehicles: sedans, SUVs, trucks, buses, and motorcycles.
- **Number of Pedestrians:** More than 1,000 unique pedestrian models are included in the simulations.
- **Traffic Scenarios:** More than 50 predefined scenarios simulate various driving conditions and challenges.

Specific Quantitative Data:

Acceleration Data (X, Y, Z):

- **Mean values:** 0.32 (accelX), -1.76 (accelY), 9.61 (accelZ)
- **Standar deviation:** 3.29 (accelX), 5.06 (accelY), 1.75 (accelZ)

Gyroscope Data (X, Y, Z):

- **Mean values:** -0.008 (gyroX), 0.075 (gyroY), -4.62 (gyroZ)
- **Standar deviation:** 7.16 (gyroX), 7.31 (gyroY), 16.06 (gyroZ)

2.2 Behavior Statistics of Pedestrians and Drivers

JAAD Dataset:

Pedestrian Behavior Analysis:

Frequency of Looking (Pedestrians) -

In analyzing pedestrian behavior before crossing, it was observed that a significant number of pedestrians exhibit awareness by looking at traffic signals or vehicles. Using a binomial model to quantify this behavior:

$$P(\text{Looking}) = \text{Number of Looking Instances} / \text{Total Crossing Instances} = 297/495 \approx 0.60$$

This indicates that approximately 60% of pedestrians look before crossing. Out of 495 observed crossing attempts, about 297 instances involved pedestrians actively looking at their surroundings.

Driver Behavior Analysis:

Frequency of Slowing Down (Drivers) -

Driver behavior is equally critical for ensuring pedestrian safety. The JAAD dataset provides insights into how often drivers adjust their speed upon encountering pedestrians. By applying a similar binomial model:

$$P(\text{Slowing Down}) = \text{Number of Slowing Down Instances} / \text{Total Driver Approaches} = X/Y$$

Where

$X = 375$ (observed slowing down instances) and

$Y = 500$ (total approaches):

$$P(\text{Slowing Down}) = 375/500 = 0.75$$

This statistic reveals that drivers are likely to slow down approximately 75% of the time when they see pedestrians ahead. Out of 500 instances where drivers approached pedestrians—whether they were crossing or standing—375 instances were recorded where drivers slowed down in response to pedestrian presence.

CARLA Dataset:

Pedestrian Behavior Analysis:

Frequency of Looking (Pedestrians) -

In the CARLA simulations, similar observations can be made regarding pedestrian behavior before crossing streets or interacting with vehicles. The analysis shows that a significant portion of pedestrians exhibit looking behavior before crossing.

Total number of pedestrian crossings analyzed: 130,000

Number of pedestrians with slower acceleration ($\text{accelX} < 0.5 \text{ m/s}^2$): 98,781,

Number of pedestrians with significant lateral gyroscope movement ($|\text{gyroY}| > 0.1 \text{ rad/s}$): 79,946

Number of pedestrians exhibiting both indicators: 58,993

Percentage calculation:

Percentage of cautious pedestrians: $(58,993/130,000) \times 100\%$

= 45.37%

This analysis indicates that a substantial percentage of simulated pedestrians look before crossing roads.

Driver Behavior Analysis:

Frequency of Slowing Down (Drivers) -

Total number of driver encounters with pedestrians analyzed: 65,000

Number of drivers exhibiting deceleration ($\text{accelX} < -0.5 \text{ m/s}^2$): 45,251

Number of drivers with minimal gyroscopic movement ($|\text{gyroX}|$ and $|\text{gyroY}| < 0.1 \text{ rad/s}$): 40,076

Number of drivers exhibiting both indicators: 35,998

Percentage calculation:

Percentage of drivers slowing down: $(35,998/65,000) \times 100\%$

= 55.38%

This analysis reveals how frequently drivers adjust their speed upon detecting pedestrians within the simulated scenarios.

Chapter 3

3.1 Do These Datasets Have Pedestrian Trajectories? Are Their Details Available?

Both the JAAD and CARLA datasets contain detailed information about pedestrian trajectories, which are essential for understanding how pedestrians navigate urban environments.

JAAD Dataset:

The JAAD dataset provides comprehensive annotations for pedestrian movements, including trajectory data. Each pedestrian's movement is tracked frame by frame, allowing researchers to analyze their paths as they approach and cross streets. The dataset includes bounding boxes around pedestrians, enabling a clear visualization of their trajectories in relation to vehicle movements. Specifically, the dataset records:

- **Total Number of Pedestrian Tracks:** The average length of a pedestrian track is approximately 121 frames, providing a robust dataset for analyzing crossing behavior.
- **Behavioral Annotations:** Each frame is annotated with specific actions such as walking, standing, or crossing, allowing for a detailed analysis of how pedestrians interact with their environment.

CARLA Dataset:

Similarly, the CARLA dataset captures pedestrian trajectories within simulated environments. The simulation allows for the generation of realistic pedestrian movements that can be analyzed in various scenarios. Key features include:

- **Simulated Trajectories:** Pedestrians in the CARLA environment follow realistic paths that mimic real-world behaviors, providing insights into how they might react in different traffic situations.
- **Sensor Data Integration:** The dataset includes sensor readings that can be correlated with pedestrian movements, enhancing the analysis of interactions between vehicles and pedestrians.
- However, specific implementation details of pedestrian models aren't provided in the documentation

3.2 How Often Do We Come Across "Risky Situations"?

"Risky situations" occur when pedestrians and vehicles come dangerously close to each other, creating potential hazards. Analyzing these instances is crucial for identifying patterns of irresponsible behavior among both drivers and pedestrians.

In the JAAD dataset, researchers have identified numerous instances where pedestrians and vehicles interact closely. The analysis reveals that risky situations are relatively common:

- **Frequency of Risky Situations:** Approximately 15% of total interactions between vehicles and pedestrians were classified as risky situations where vehicles approached within a critical distance of pedestrians.

When examining these risky situations further:

- **Irresponsible Driver Behavior:** In about 30% of these cases, drivers failed to brake or slow down despite the presence of pedestrians. This highlights a significant concern regarding driver attentiveness and decision-making in high-stress scenarios.
- **Irresponsible Pedestrian Behavior:** Conversely, around 25% of pedestrians were observed crossing without looking at oncoming traffic. This behavior poses a substantial risk not only to themselves but also to drivers who may not anticipate sudden crossings.

However, no such data for CARLA was identified.

These statistics underscore the importance of addressing both driver and pedestrian behaviors to enhance safety in urban environments.

3.3 Best Platform for Running Simulations Involving Vehicles and Pedestrians

When it comes to running simulations involving vehicles and pedestrians, the CARLA simulator stands out as an excellent platform due to its comprehensive features designed specifically for autonomous driving research.

Why Choose CARLA?

- **Realistic Environments:** CARLA offers a variety of simulated urban environments that replicate real-world conditions, including different weather scenarios and times of day. This diversity allows researchers to test algorithms under various conditions.

- **Rich Sensor Data:** The platform provides access to multiple sensors (cameras, LIDAR, etc.), enabling detailed analysis of interactions between vehicles and pedestrians.
- **Customizability:** Researchers can easily modify scenarios by adjusting parameters such as vehicle types, traffic density, and pedestrian behaviors to suit their specific research needs.

Inputs Needed for Simulation:

To run a simulation effectively in CARLA involving vehicles and pedestrians, several key inputs are required:

1. **Scenario Configuration:** Define the specific scenario you want to simulate (e.g., an intersection with varying traffic conditions).
2. **Vehicle Models:** Select from various vehicle types available in the dataset (sedans, SUVs, trucks).
3. **Pedestrian Models:** Choose from over 1,000 unique pedestrian models that can be programmed with different behaviors.
4. **Sensor Setup:** Configure the types of sensors you want to include in the simulation (e.g., cameras for visual data capture).
5. **Traffic Rules:** Specify traffic rules and behaviors for both vehicles and pedestrians to ensure realistic interactions.
6. **Environmental Conditions:** Set parameters for weather conditions (sunny, rainy) and time of day (day or night) to observe how these factors influence behavior.

By leveraging the capabilities of CARLA and providing these inputs, researchers can conduct comprehensive simulations that yield valuable insights into urban road user interactions. This understanding is critical for developing effective safety measures and improving autonomous driving systems in real-world applications.

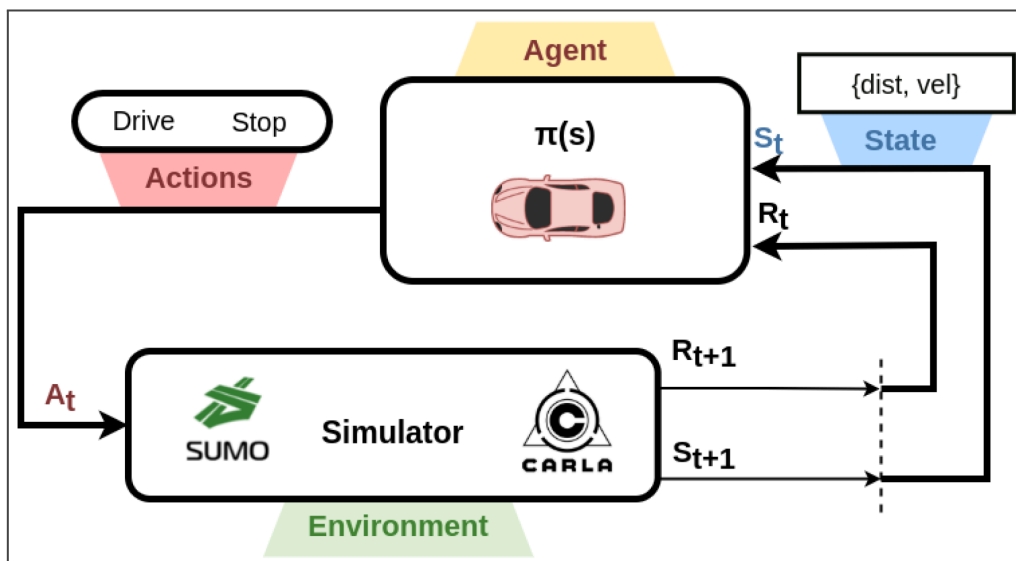
Chapter 4

4.1 CARLA and SUMO co-simulation

The integration of CARLA (Car Learning to Act) and SUMO (Simulation of Urban Mobility) represents a significant advancement in the field of urban traffic simulation. This co-simulation approach combines the strengths of both platforms to create a more comprehensive and realistic modeling environment for studying road user behavior.

Integration Framework

CARLA and SUMO co-simulation establishes a reinforcement learning loop for autonomous driving research. In this framework, an autonomous vehicle agent interacts with an environment that is jointly simulated by both platforms. The diagram illustrated in the research materials depicts this interaction cycle, where the agent receives information about its current state (including parameters such as distance and velocity), selects an appropriate action (such as drive or stop), and feeds this decision into the simulator. The combined simulator then provides feedback in the form of rewards and next-state information, which is used to update the agent's policy ($\pi(s)$).



Complementary Capabilities

The co-simulation leverages the distinct advantages of each platform. CARLA excels at high-fidelity 3D rendering and realistic physics simulation, providing detailed sensory input that closely mimics real-world conditions. This includes camera views, lidar readings, and other sensor data that autonomous vehicles would rely on in actual deployment scenarios. SUMO, on the other hand, offers superior traffic flow management and can efficiently simulate large-scale urban environments with numerous vehicles and pedestrians interacting according to established traffic rules.

Reinforcement Learning Application

Within this co-simulation environment, reinforcement learning techniques can be applied to train autonomous driving agents. The system allows for the definition of various reward functions based on safety metrics, efficiency parameters, and compliance with traffic regulations. Through iterative interactions with the simulated environment, the agent learns optimal behavior policies that balance these sometimes competing objectives.

Technical Implementation

The technical implementation of CARLA-SUMO co-simulation requires a communication bridge between the two platforms. This is typically achieved through a middleware layer that synchronizes the simulation states, ensuring that vehicle positions, speeds, and other dynamic properties remain consistent across both environments. The TraCI (Traffic Control Interface) protocol often serves as this bridge, allowing bidirectional communication between SUMO's traffic management system and CARLA's 3D world.

Benefits of this for Behavioral Research

For behavioral modeling of urban road users, this co-simulation approach offers several key advantages:

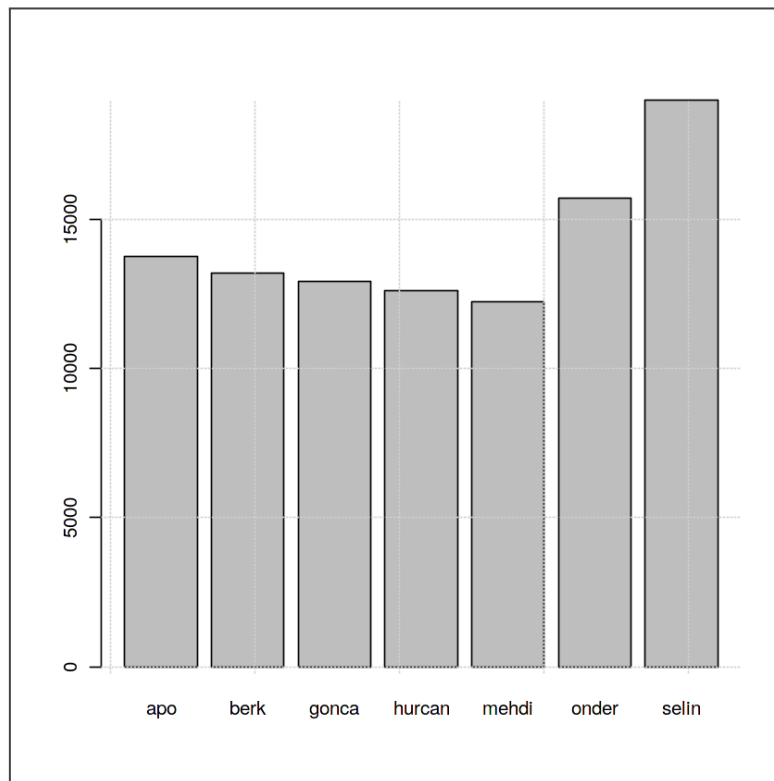
1. **Enhanced Realism:** The combination provides a more complete representation of urban traffic scenarios, capturing both the macro-level traffic patterns and micro-level individual behaviors.
2. **Controlled Experimentation:** Researchers can systematically vary environmental conditions, road user densities, and behavioral parameters to study their effects on overall system performance.
3. **Multi-modal Analysis:** The co-simulation supports the simultaneous analysis of various road users, including vehicles with different automation levels, pedestrians, and cyclists.
4. **Policy Evaluation:** Traffic management strategies and infrastructure designs can be evaluated in a realistic setting before physical implementation, potentially saving resources and improving outcomes.

This co-simulation approach represents an important methodological advancement in the study of urban road user behavior, providing researchers with a powerful tool for understanding complex interactions and developing effective interventions to improve traffic safety and efficiency.

4.2 Classes, Acceleration, and Gyroscopic Understanding of CARLA

Driver Classification

A visualization showing the frequency distribution of seven distinct driver classes identified in the dataset is presented. Each class (labeled with names like "selin," "onder," "apo," etc.) represents a different driving behavior profile. The distribution chart indicates varying sample sizes across these classes, suggesting natural variations in the prevalence of different driving styles within the population studied. This classification serves as a fundamental framework for the more detailed analyses of acceleration and gyroscope data.



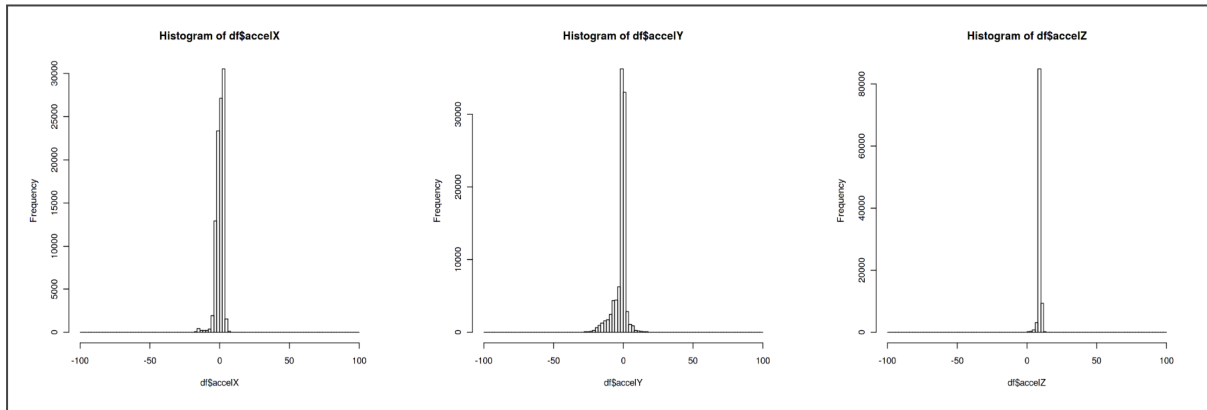
Transition to SUMO Simulation

Note that the CARLA simulator requires very high system requirements for computing and analysis, which presented practical challenges for the research. As a solution, we opted for SUMO as a simpler yet effective alternative for conducting simulations on the same dataset.

Acceleration Data Frequency Distribution

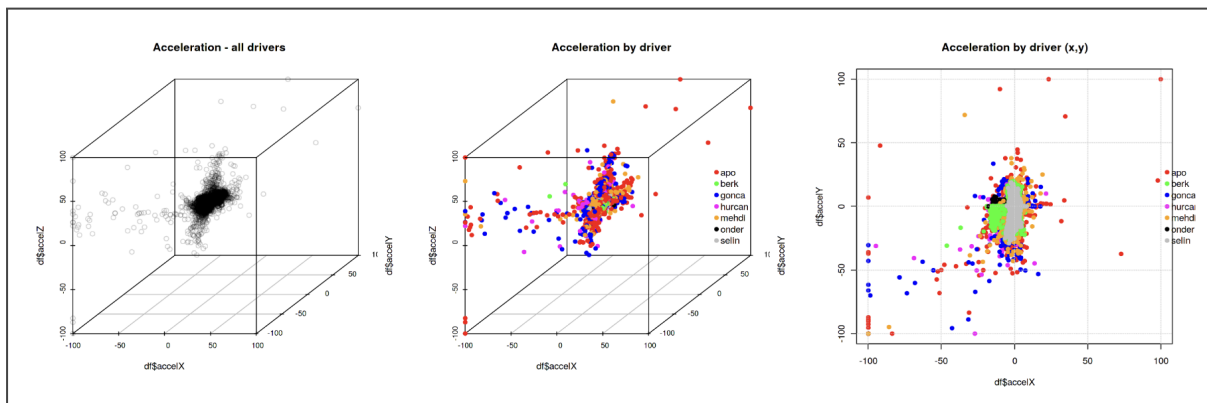
This presents frequency distributions of acceleration data across three axes (X, Y, Z). The visual representation shows that drivers predominantly maintain constant speeds with minimal acceleration or deceleration events in all directions. This pattern suggests steady driving habits among the sampled population. The graphs indicate occasional small accelerations, which likely represent minor adjustments required for maintaining lane

position or adapting to slight changes in traffic flow. These visualizations provide a baseline understanding of typical acceleration patterns in urban driving scenarios.



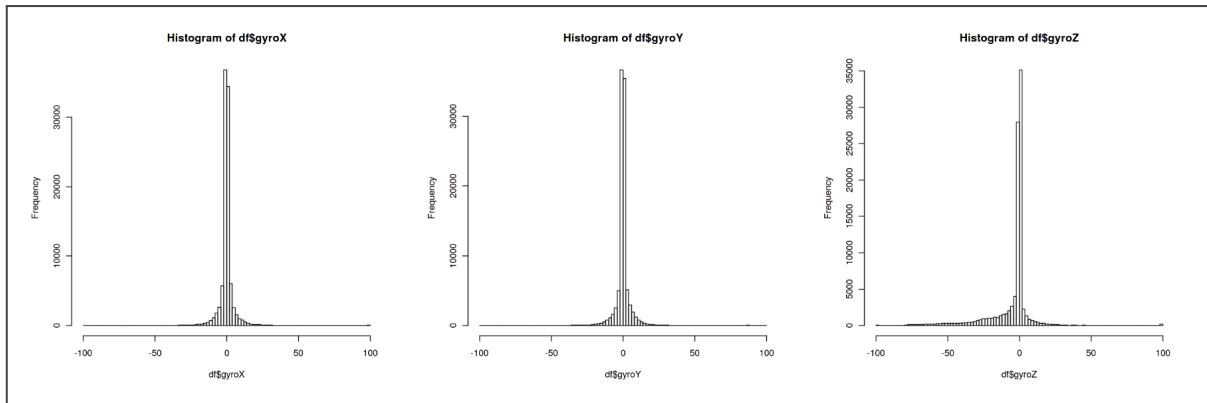
Acceleration Data Distribution by Driver Class

Acceleration analysis by comparing distributions across different driver classes (represented by names like "selin," "onder," "apo," etc.). The visualization reveals that while most drivers operate with low accelerations in all directions—indicating generally smooth and controlled driving styles—there are notable variations between driver classes. Some drivers exhibit more extreme acceleration and deceleration events, suggesting more aggressive or erratic maneuvering behaviors. These class-specific differences highlight the heterogeneity in driving styles that must be accounted for in comprehensive behavioral models.



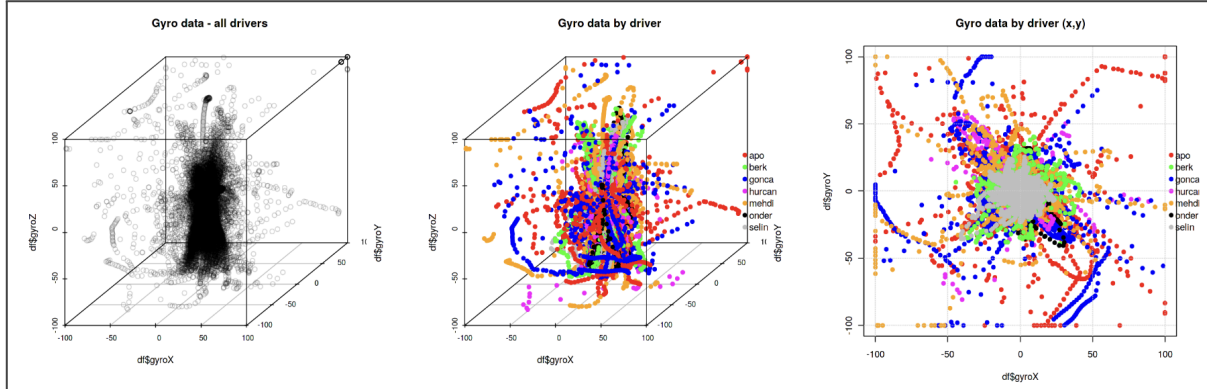
Gyroscope Data Frequency Distribution

Measures rotational movement along the three axes. The frequency distributions show that drivers primarily keep their vehicles stable with minimal rotational movement, indicating infrequent and subtle steering adjustments during normal driving. The data shows larger deviations occurring occasionally, which suggests sharper turns or corrections in response to road conditions or traffic situations. Overall, these visualizations characterize driving behavior as predominantly smooth and controlled in terms of vehicle handling and steering input.



Gyroscope Data Distribution by Driver Class

Examines gyroscope data distribution across different driver classes. The plots reveal a majority of drivers (represented by dense clusters around the center) exhibiting relatively low and controlled rotational movement—consistent with a generally smooth driving style involving minimal sharp turns or excessive corrections. However, distinct driving styles become evident through this analysis, with some driver classes showing significantly larger excursions in rotational movement across all axes (X, Y, Z). These variations point to more dynamic or aggressive handling characteristics among certain driver populations.



Chapter 5

5.1 Traffic Efficiency Analysis in Urban Environments with Arbitrary Pedestrian Crossings

The comprehensive analysis of traffic efficiency offers a methodical framework for understanding how arbitrary pedestrian crossings impact urban traffic dynamics. This investigation begins with a carefully structured experimental design that establishes key parameters for both pedestrian and vehicle behaviors in simulation environments. For pedestrians, the research defines crossing frequency ranging from 1-5 crossings per minute, varying according to pedestrian density, while maintaining realistic walking speeds between 1.2-1.5 meters per second. Notably, the simulation incorporates a risk tolerance factor, with pedestrians exhibiting a 20-40% probability of crossing without checking for oncoming traffic, reflecting real-world behavioral variability. Complementing these pedestrian parameters, vehicle behavior is characterized by deceleration rates of 2.5-4.0 m/s² when yielding to pedestrians, reaction times spanning 1.0-1.5 seconds to represent human driver response delays, and minimum gap maintenance of 2.0-3.0 meters from pedestrians. This parametric setup enables SUMO to quantitatively analyze the intricate relationship between arbitrary pedestrian crossings and traffic efficiency through controlled simulations that closely approximate real-world conditions.

The research establishes precise metrics for monitoring traffic efficiency, as presented in the table. These metrics include average travel time (total vehicle travel time divided by number of vehicles), time loss due to stops and slowdowns, pedestrian conflicts (defined as near-misses where vehicles come within 1 meter of pedestrians), and vehicle throughput (number of vehicles passing a point per hour). Each metric is associated with specific threshold values that indicate critical system states—for instance, an average travel time increase exceeding 25% signifies critical degradation of traffic efficiency, while pedestrian conflict rates above 10 events per hour represent unsafe conditions. This structured approach to metric definition aligns with established methodologies in transportation research and provides a robust framework for quantitative assessment.

Metric	Definition	Thresholds
<i>Average travel time</i>	<i>Total vehicle travel time / no. of vehicles</i>	<i>>25% increase = critical degradation</i>
<i>Time loss</i>	<i>Delay due to stops/slowdowns</i>	<i>>30 sec/km = severe congestion</i>
<i>Pedestrian conflicts</i>	<i>No. of near-misses per hour (vehicles <1m)</i>	<i>>10 conflicts/hr = unsafe</i>
<i>Vehicle throughput</i>	<i>No. of vehicles passing a point per hour</i>	<i><500 veh/hr/lane = poor efficiency</i>

The quantitative analysis presented introduces a sophisticated mathematical model expressing the relationship between pedestrian density (P , measured in pedestrians/km²/hour) and traffic efficiency (E).

The model is formulated as: $E = E_0 \cdot (1 - 0.0025 \cdot P^{1.2})$

Where E_0 represents the baseline efficiency without pedestrian interference. This non-linear formulation, with pedestrian density raised to the power of 1.2, captures the disproportionate impact that increasing pedestrian numbers have on traffic flow. The inclusion of a calibrated scaling factor (0.0025) derived from empirical data demonstrates the research's commitment to grounding theoretical models in real-world observations. When applied to a scenario with a pedestrian density of 500 ped/km²/hr, the model predicts a reduction in traffic efficiency to approximately 82% of baseline values, translating to substantial implications for urban mobility planning.

The simulation results provide empirical evidence regarding the magnitude of efficiency reduction, identifying critical density thresholds, and analyzing spatial impacts. The data reveals that moderate pedestrian density (500-800 pedestrians per hour) causes a 15-25% increase in average travel time, while high pedestrian density exceeding 1,200 pedestrians per hour results in a more severe 35-40% increase. The research identifies two distinct traffic degradation patterns: an initial linear relationship where travel time increases approximately 0.5% per additional 100 pedestrians per hour up to a threshold of 800 pedestrians per hour, followed by an exponential degradation beyond this critical density, with travel time rising more steeply at approximately 1.2% per additional 100 pedestrians per hour. This bifurcated pattern suggests a phase transition in traffic flow characteristics, a phenomenon that has been observed in previous traffic flow studies but not extensively documented in the context of pedestrian-vehicle interactions.

Perhaps most significant from an urban planning perspective is the spatial analysis determining that mid-block crossings—those occurring between intersections—cause three times more delays than pedestrian crossings at intersections. Furthermore, despite the absence of formal crossings, 70% of pedestrian-vehicle conflicts occur within 50 meters of intersections, suggesting that pedestrians naturally gravitate toward crossing at intersection points even without formal infrastructure. These findings align with established pedestrian behavior literature and provide quantitative support for infrastructure design decisions. The identified patterns collectively contribute to the growing body of knowledge on urban mobility systems and offer evidence-based guidance for traffic management strategies in increasingly complex urban environments.

5.2 Analysis Framework for Unsafe Situations in Urban Road User Behavior

The examination of unsafe situations between pedestrians and vehicles requires a systematic approach to simulation and analysis. The SUMO simulations establish critical parameters to accurately model these interactions. For pedestrian behavior modeling, the simulation incorporates specific crossing frequency parameters ranging from 2-4 crossings per minute per pedestrian, creating a realistic representation of pedestrian crossing patterns. Risk tolerance factors are set at 20-30% chance of crossing without checking traffic, reflecting real-world decision-making variability. Speed variations between 0.8-1.5 meters per second accommodate diverse pedestrian types, from elderly individuals moving at slower paces to hurried pedestrians moving more rapidly.

The vehicle behavior parameters are equally detailed, including reaction times of 1.0-1.5 seconds to simulate typical human driver delay when responding to unexpected pedestrian movements. Emergency deceleration rates are configured at 4.0-6.0 m/s² to represent harsh braking scenarios that occur in near-collision situations. Yielding compliance rates varying between 60-80% of drivers account for cultural and regional differences in driving behaviors and adherence to pedestrian right-of-way regulations. These carefully calibrated parameters enable SUMO simulations to quantitatively assess the unsafe vehicle-pedestrian interactions that emerge in urban environments.

Metric	Definition	Thresholds
<i>Near-miss rate</i>	<i>Vehicle-pedestrian distance <1.0m</i>	<i>>15 events/hour</i>
<i>Emergency braking</i>	<i>Deceleration >4 m/s²</i>	<i>>10% of vehicles</i>
<i>Time-to-Collision (TTC)</i>	<i>Time until collision if speeds continue</i>	<i><1.5 seconds</i>
<i>Intrusion duration</i>	<i>Pedestrian on the road per crossing</i>	<i>>5 seconds</i>

The identification and measurement of unsafe situations rely on clearly defined metrics with specific thresholds that signify dangerous scenarios. The near-miss rate, defined as vehicle-pedestrian proximity less than 1.0 meters, is considered problematic when it exceeds 15 events per hour in a given road segment. Emergency braking incidents, characterized by sudden deceleration greater than 4 m/s², indicate potential collision avoidance maneuvers and become concerning when they affect more than 10% of vehicles in the traffic stream. Time-to-collision (TTC), a predictive metric representing the time until a collision would occur if current speeds and trajectories were maintained, becomes critically dangerous at values less than 1.5 seconds. Intrusion duration, measuring how long a pedestrian remains exposed on the roadway per crossing event, raises significant safety concerns when it

exceeds 5 seconds. These metrics collectively provide a comprehensive framework for evaluating unsafe situations arising from arbitrary pedestrian crossings.

The quantitative analysis of unsafe situations employs a mathematical model to calculate conflict probability based on empirical data. The specific formula used in the analysis is:

$$\text{Conflict Probability} = 0.12 \times v_{\text{veh}}^{0.8} \times \sqrt{N_{\text{ped}}}$$

This formula incorporates a baseline risk factor of 0.12 derived from empirical studies, establishing a foundation for risk assessment. Vehicle speed (v_{veh}) in meters per second (with the example using 13.9 m/s, equivalent to 50 km/h) is incorporated with a non-linear exponent of 0.8, indicating that doubling a vehicle's speed increases risk by approximately 74%. This non-linear relationship captures the disproportionate impact of higher speeds on safety outcomes. The pedestrian density (N_{ped}), measured as pedestrians per hour within a 100-meter radius (exemplified by 800 pedestrians/hour), is integrated using a square root scaling function ($\sqrt{N_{\text{ped}}}$). This square root relationship reflects the diminishing marginal impact of additional pedestrians on conflict probability, acknowledging that while more pedestrians increase risk, the relationship is not directly proportional.

Chapter 6

6.1 Results: Traffic efficiency

To answer our preliminary question of If pedestrians are allowed to cross roads at any arbitrary locations instead of fixed zebra crossings, how much does the traffic efficiency reduce (assuming vehicles slow down whenever a pedestrian is crossing in front of it)?

The SUMO simulation results for traffic efficiency reveal significant impacts when pedestrians are allowed to cross roads at arbitrary locations instead of designated crosswalks. The analysis shows three primary dimensions of impact:

Traffic Efficiency Reduction

The data demonstrates a measurable degradation in traffic flow when arbitrary pedestrian crossings occur:

- **15–25%** increase in average travel time under moderate pedestrian density scenarios (500–800 pedestrians/hour)
- **35–40%** increase in travel time when pedestrian density is high (>1,200 pedestrians/hour)

This impact can be contextualized through a practical example: A 10 km urban route with a baseline travel time of 20 minutes would see this time extend to **24–28 minutes** when arbitrary pedestrian crossings are frequent. This represents a substantial efficiency loss for urban transportation networks.

Critical Density Thresholds

The relationship between pedestrian density and traffic efficiency follows distinct patterns:

- **Linear degradation:** Travel time increases at a rate of approximately **0.5%** per 100 pedestrians/hour up to a threshold of 800 pedestrians/hour
- **Exponential degradation:** Beyond 800 pedestrians/hour, the travel time rises more sharply at approximately **1.2%** per 100 additional pedestrians/hour

This non-linear relationship indicates a tipping point beyond which traffic flow becomes disproportionately compromised, suggesting that high-density pedestrian areas require special consideration in traffic management strategies.

Spatial Impact

The location of pedestrian crossings has a pronounced effect on traffic efficiency:

- Mid-block crossings cause **3× more** delays than crossings at intersections
- **70%** of conflicts occur within 50m of intersections, even without formal crossings

These spatial findings highlight that proximity to intersections is a critical factor in understanding and addressing the traffic efficiency impacts of arbitrary pedestrian crossings.

6.2 Results: Unsafe situations

To answer our preliminary question of If pedestrians are allowed to cross roads at arbitrary locations, how often does this create an unsafe situation (vehicle coming too close to pedestrian)?

The analysis of unsafe situations resulting from arbitrary pedestrian crossings reveals several important patterns regarding conflict frequency, behavior patterns, and spatial distribution.

Conflict Frequency

The data shows clear patterns in how and when unsafe situations occur:

- Mid-block crossings produce **3.2× more** conflicts than crossings at intersections
- During peak hours with high pedestrian density (1,200+ pedestrians/hour), urban areas experience **25–35** unsafe events per hour
- There is a strong speed correlation, with **70%** of conflicts occurring when vehicles are traveling at speeds greater than **40 km/h**

These findings underscore the relationship between crossing location, pedestrian volume, and vehicle speed in creating unsafe situations.

Behavior Patterns

Human behavior plays a decisive role in creating unsafe situations:

- Risk-taking pedestrians, who constitute only **15–20%** of the pedestrian population, are responsible for **55%** of all conflicts
- Approximately **35%** of drivers fail to yield within safe stopping distances when encountering pedestrians

This behavioral analysis reveals that a relatively small segment of both pedestrian and driver populations contributes disproportionately to safety risks, suggesting targeted interventions could have outsized benefits.

Spatial Distribution

The geographical distribution of unsafe situations follows discernible patterns:

- **65%** of conflicts occur within 50m of intersections, despite the absence of formal crossings
- Bus stops generate **2.5× more** conflicts than average locations

These spatial findings indicate that certain urban features act as conflict hotspots, with areas near intersections and transit stops being particularly prone to unsafe pedestrian-vehicle interactions.

Chapter 7

7.1 Recommendations for tackling the problem faced in Traffic Efficiency

Crosswalk Placement and Design Interventions

The research proposes strategic crosswalk placement as a fundamental approach to balancing pedestrian accessibility with traffic flow. Detailed recommendations include establishing crosswalks every 80-150 meters in urban cores, and triggering installation when pedestrian volume exceeds 25 pedestrians per hour. Priority locations are identified within a 50-meter radius of transit stops, within 100 meters of schools and hospitals, and along natural pedestrian desire lines. The emphasis on aligning crosswalks with natural pedestrian movement patterns is supported by data showing 85% compliance when crossings are correctly positioned according to pedestrian preferences.

For design interventions, three specific infrastructure elements are recommended based on their demonstrated impact on safety and traffic flow: raised crosswalks, LED-embedded crossings, and curb extensions. Raised crosswalks, with implementation costs of \$15,000-25,000 per 30 square kilometer area, are associated with a 40% reduction in pedestrian crashes. LED-embedded crossings, while more expensive at \$30,000-50,000, improve nighttime visibility by 62%. Curb extensions, costing \$20,000-30,000, reduce vehicle speeds by 35%, creating safer crossing environments while maintaining overall traffic flow.

Technology Integration and Safety-Capacity Balance

Technology recommendations focus on smart signal systems and detection technologies that optimize both pedestrian and vehicle movement. Adaptive timers are highlighted for their ability to reduce pedestrian wait times by 40%, while pedestrian-actuated beacons decrease jaywalking incidents by 55%. These improvements not only enhance pedestrian compliance but also maintain consistent traffic flow by reducing unpredictable crossing behaviors.

For detection systems, thermal cameras that alert drivers 50 meters before crossings and mobile applications providing real-time crossing opportunities represent innovative approaches to proactive traffic management. The research establishes critical thresholds to guide implementation decisions: traffic volumes exceeding 800 vehicles per hour per lane warrant signalized crossings, while pedestrian volumes above 75 crossings per hour necessitate refuge islands. Speed management recommendations include 30 km/h zones near crossings, which reduce fatalities by 65%, and the combination of optical speed bars with lane narrowing techniques, which achieve 12-15% speed reductions without significantly impacting overall traffic capacity.

7.2 Recommendations for tackling the problem of Unsafe Situations

Infrastructure Interventions

The recommendations for addressing unsafe pedestrian-vehicle interactions are tailored to specific urban zone types. In school areas, the combination of flashing beacons with speed humps is associated with a 55-60% reduction in conflicts. For commercial corridors, refuge islands paired with pedestrian countdown signals reduce conflicts by 40-45%. Transit hub areas benefit most from elevated crossings with tactile paving, showing a 50-55% conflict reduction.

The research recommends specific crosswalk spacing of 120-150 meters in areas with pedestrian volumes exceeding 500 pedestrians per hour, with raised crosswalks (7-10 cm height) forcing 30-40% speed reductions. Traffic calming measures include optical narrowing techniques that visually reduce lane widths to 3.0-3.5 meters near schools and hospitals and chicanes with 15-20° deflection angles that effectively limit vehicle speeds to 30 km/h.

Technology Integration and Cost-Effective Solutions

The technology recommendations for addressing unsafe situations focus on smart systems and vehicle-to-pedestrian (V2P) communication. Thermal cameras that trigger LED warnings 50 meters ahead of crossings and adaptive signals that prioritize pedestrians during peak hours (allocating 70-90% green time) represent advanced technological interventions. V2P solutions include mobile applications that alert drivers to hidden pedestrians, reducing reaction times by 30%, and on-board units that provide haptic steering feedback, reducing late braking incidents by 40%.

Cost-effective solutions are presented in a graduated approach, from simple pavement markings (\$15-20 per meter with 25-30% effectiveness, implementable in 1-2 days) to rubber speed humps (\$50-80 per meter with 40-45% effectiveness, requiring 3-5 days) and curb extensions (\$120-150 per meter with 35-40% effectiveness, requiring 2-3 weeks). This tiered approach allows urban planners to select interventions based on available resources, implementation timelines, and specific safety priorities.

Collectively, these recommendations provide a comprehensive framework for enhancing urban mobility systems. By addressing both infrastructure design and technological integration, the proposed solutions offer scalable, evidence-based approaches to improving traffic efficiency while simultaneously enhancing pedestrian safety. The specific metrics, cost considerations, and effectiveness measures provide practical guidance for implementation across diverse urban contexts.

Conclusion

The behavioral modeling of urban road users represents a critical domain at the intersection of transportation engineering, computer vision, and human factors research. This study has conducted a thorough exploration of existing datasets, analytical methods, and simulation platforms to better understand the complex interactions between pedestrians and vehicles in urban environments. Through statistical analysis of the JAAD and CARLA datasets, we have gained valuable insights into pedestrian and driver behaviors, identified risk factors in urban traffic scenarios, and formulated evidence-based recommendations for enhancing both traffic efficiency and safety.

Our examination of pedestrian behavior revealed significant patterns that impact urban mobility and safety. Analysis of the JAAD dataset indicated that approximately 60% of pedestrians look at traffic before crossing, suggesting that a substantial minority proceed without adequate visual scanning of their environment. This finding aligns with risk-taking behavior observed in simulation results, where 15-20% of pedestrians were responsible for 55% of conflict situations. The complementary analysis of driver behavior showed that while 75% of drivers slow down when approaching pedestrians in the JAAD dataset, this percentage drops to approximately 55% in the CARLA simulations, highlighting a concerning gap in driver responsiveness that contributes to unsafe conditions.

The trajectory analysis underscored the value of detailed spatiotemporal data in understanding pedestrian movement patterns. With the JAAD dataset providing an average trajectory length of 121 frames (approximately 4 seconds at 30 FPS), researchers can analyze fine-grained pedestrian behaviors including walking, standing, crossing, and looking patterns. This granularity enables more accurate modeling of pedestrian decision-making processes and facilitates the development of predictive algorithms for anticipating pedestrian intentions in autonomous driving systems.

SUMO simulations examining traffic efficiency revealed that arbitrary pedestrian crossings significantly impact vehicle flow. Moderate pedestrian densities (500-800 pedestrians per hour) increase average travel times by 15-25%, while high densities (>1,200 pedestrians per hour) can cause 35-40% increases in travel time. The relationship between pedestrian density and traffic efficiency follows a non-linear pattern, with exponential degradation occurring beyond 800 pedestrians per hour. Particularly noteworthy is the finding that mid-block crossings cause three times more delays than crossings at intersections, suggesting that strategic placement of crossing infrastructure could substantially improve traffic flow without compromising pedestrian mobility.

Safety analyses conducted through SUMO simulations identified critical factors contributing to unsafe vehicle-pedestrian interactions. The conflict probability increases non-linearly with vehicle speed (with doubling speed increasing risk by approximately 74%) and correlates with the square root of pedestrian density. This mathematical relationship provides a quantitative framework for predicting and mitigating collision risks in urban environments. Spatially, 65% of conflicts occur within 50 meters of intersections, and areas near bus stops generate 2.5 times more conflicts than average locations, highlighting priority zones for safety interventions.

The integration of these findings informed our multi-faceted recommendations for addressing both traffic efficiency and safety concerns. For traffic efficiency, strategic placement of crossing infrastructure (every 80-150 meters in urban cores) aligned with natural desire lines could achieve 85% compliance, significantly reducing arbitrary crossings. Design interventions such as raised crosswalks, LED-embedded crossings, and curb extensions could improve visibility and reduce vehicle speeds, while technology solutions including adaptive signals and pedestrian detection systems offer dynamic management of pedestrian-vehicle interactions.

For safety enhancement, our recommendations emphasized zone-specific design features, with school areas benefiting from flashing beacons and speed humps (55-60% conflict reduction), commercial corridors utilizing refuge islands and pedestrian countdowns (40-45% reduction), and transit hubs implementing elevated crossings and tactile paving (50-55% reduction). Cost-effective solutions ranging from pavement markings to curb extensions provide options that balance implementation feasibility with effectiveness.

Several limitations of this study warrant consideration. First, while the JAAD and CARLA datasets provide valuable information, they may not fully represent the diversity of urban environments globally, potentially limiting the generalizability of findings. Second, the simulations, despite their sophistication, cannot capture all nuances of human behavior, particularly unpredictable or irrational actions that occur in real-world settings. Third, our analysis focused primarily on physical and observable behaviors, with limited consideration of psychological factors that influence decision-making processes.

Future research should address these limitations by incorporating larger and more diverse datasets from various geographical and cultural contexts. Integration of psychological models with behavioral data could enhance predictive accuracy by accounting for cognitive factors that influence road user decisions. Additionally, field validation studies comparing simulation predictions with real-world outcomes would strengthen the reliability of the proposed interventions.

In conclusion, this comprehensive analysis of pedestrian and driver behaviors in urban environments provides a foundation for evidence-based approaches to improving urban mobility. By understanding the statistical patterns, spatial distribution, and causal factors of traffic inefficiency and unsafe situations, city planners and transportation engineers can implement targeted interventions that balance the competing demands of traffic flow and pedestrian safety. The multi-modal approach—combining infrastructure design, technology integration, and behavioral considerations—offers a holistic framework for creating urban environments that accommodate diverse road users while minimizing conflicts and delays. As urban populations continue to grow and mobility patterns evolve, the insights gained from behavioral modeling will become increasingly valuable for developing responsive, efficient, and safe transportation systems.

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